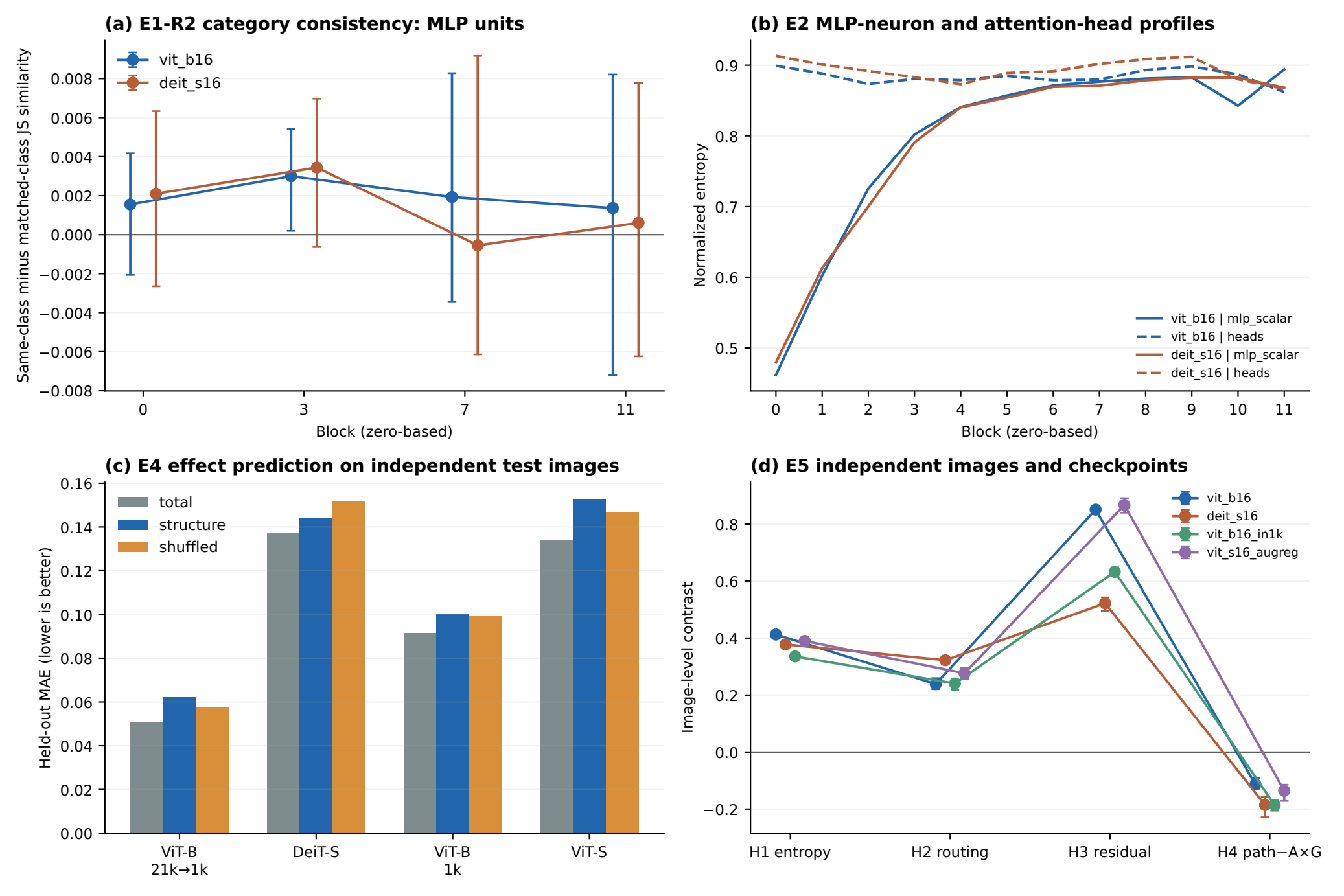
(a) E1-R2 category consistency: MLP units
Same-class minus matched-class JS similarity
vit_b16
deit_s16
0.008
0.006
0.004
0.002
0.000
−0.002
−0.004
−0.006
−0.008
0
3
7
11
Block (zero-based)
(b) E2 MLP-neuron and attention-head profiles
Normalized entropy
0.9
0.8
0.7
0.6
0.5
vit_b16 | mlp_scalar
vit_b16 | heads
deit_s16 | mlp_scalar
deit_s16 | heads
0
1
2
3
4
5
6
7
8
9
10
11
Block (zero-based)
(c) E4 effect prediction on independent test images
Held-out MAE (lower is better)
total
structure
shuffled
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
ViT-B
21k→1k
DeiT-S
ViT-B
1k
ViT-S
(d) E5 independent images and checkpoints
Image-level contrast
vit_b16
deit_s16
vit_b16_in1k
vit_s16_augreg
0.8
0.6
0.4
0.2
0.0
−0.2
H1 entropy
H2 routing
H3 residual
H4 path−A×G